Enhancing Aging-in-Place with AI-Personalized Recommendations and Home-Oriented Smart Technologies

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Background

- By 2030, nearly 1 in 5 Americans (almost 73 million) will be aged 65 or older, which represents approximately 20% of the U.S. population[1].
- Roughly 77% to 90% of older adults express a preference to "age in place"- remaining in their own homes or communities as they grow older[1].
- However, mobility and cognitive challenges are common among this demographic, with falls and medication errors being the leading causes of fatal and severe non-fatal injuries among older adults.
- Home-Oriented Smart Technologies (HOST), such as fall detectors or automated pill dispensers, offer meaningful support for older adults' safety, independence, and quality of life.

Objectives/Aims

1. Design and validate an AI-Driven platform for personalized HOST recommendations.
2. Develop a modular demonstration kit for fall prevention and medication reminder HOST training.
3. Evaluate feasibility and acceptability of the integrated AI-Kit Intervention

Methodology

Scoping Review

- A comprehensive scoping review is conducted to systematically map existing literature to identify the types of HOSTs used for aging in place, target populations, implementation settings, outcomes evaluated, key challenges, and gaps in the evidence base.

Needs Assessment

- A needs assessment is performed to identify key challenges and priorities during day-to day activities for older adults.
- Focus groups composed of older adults (age 65+) and caregivers were conducted to identify common trends of living difficulties, risks, current mitigation strategies, and willingness to adult new technologies.

Workshops

- Workshops are held to present the modular demonstration kit to older adults and caretakers, demonstrating examples of HOST, their setup, and usability.

Modular Demonstration Kit Overview

- The modular demonstration kit serves as a hands-on training platform that simulates a smart home environment tailored for older adults and caregivers.
- Designed for portability and adaptability, the kit features modular components that can be reconfigured to showcase various smart home technologies related to fall prevention, medication management, and daily living assistance.
- It enables users to interact with real-world smart devices in a controlled setting, supporting training on how to monitor, control, and respond to health and safety-related events in the home.
- The kit is a wooden, trifold demonstration unit featuring hinged pegboard panels for flexible device mounting, concealed wiring for safety and realism, and a compact, portable design that allows for easy transport and reconfiguration in various training and outreach settings.
- Progress includes the completed frame design, partial wiring, device integration plan, and initial training scenarios. Next steps involve building the prototype, testing, feedback collection, and iterative refinement to optimize usability and effectiveness.

Needs Assessment Overview

- **Demographics & Living Situation**
  - Participants identify age range (65+), living arrangement, and type of residence
  - Helps determine how living environments influence smart technology needs
- **Technology Use & Experience**
  - Assesses comfort level with smartphones, tablets, computers, and smart devices
- **Mobility & Safety**
  - Captures use of assistive devices (e.g., walker, cane, grab bars)
  - Evaluates device effectiveness and common challenges (e.g., bulky, hard to use)
- **Medication Management**
  - Identifies difficulties in remembering doses, reading labels, or opening bottles
  - Explores effects of missed doses and associated stress or health issues
- **Daily Living Activities**
  - Identifies physical or cognitive barriers and safety concerns in home areas
  - Reviews effectiveness of tools used for daily tasks
- **Smart Technology Familiarity**
  - Checks prior exposure and current use of smart tech (e.g., smart speakers, phones)
  - Explores reasons for non-use (e.g., too complex, privacy, cost, no guidance)

Results

- Initial flyer for needs assessment has been completed
- Created a model for the demo kit
- Finalized purchase of supplies for the demo kit
- Currently researching various smart home devices for older adults. Devices include fall detection systems, medication reminders, voice assistants, and home automation tools
- Team is still in the working stages of organizing and analyzing findings
- Currently working on a systematic review paper for smart me technology on older adults.
- Progress is steady as we move toward finalizing materials and recommendations

Discussion

- Supporting aging-in-place through smart technology requires a proactive, user-centered approach that integrates both education and accessible device design.
- The planned modular demonstration kit aims to provide a flexible, hands-on platform for introducing older adults and caregivers to smart home technologies that promote safety, independence, and wellness.
- Significant effort has been devoted to researching and curating suitable smart devices, with a focus on usability, reliability, and alignment with the needs of older adults in areas such as fall prevention, medication management, and daily living support.
- A comprehensive needs assessment tool is in development and will be used to gather feedback on technology use, challenges, and preferences, which will guide future device selection and training scenarios.
- As the project progresses, iterative design, stakeholder engagement, and field testing will be critical to ensure the final kit and training content are effective, acceptable, and impactful for the target population.

[1][https://pmc.ncbi.nlm.nih.gov/articles/PMC8142240/]